

Lecture 11 and 12

- short-run supply with perfect competition
- (maybe) cost functions
- long-run supply
- LR and SR equilibrium
- monopoly

Recap

- so far, we learned how individuals maximize utility to come up with their demand functions
 - and then, focusing on a single good, how to aggregate across consumers to find **market demand**
- we'll now do the production analog: how do firms maximize profits to come up with their supply curves
 - and then, how to aggregate across firms to find **market supply**
- the only slight wrinkle: the profit-maximizing strategy depends on how many competitors a firm has
- today, we'll look at one extreme, known as **perfect competition aka price-taking firms**

Profit Maximization

- in principle, companies may be private (“for profit”), public (e.g. run by the government), or nonprofit
- this being an economics course, we will focus on the first type, private companies
- maintained assumption for this course:
 - **a company’s goal is to maximize profits**

Price-Taking Firms

- we'll start with the simplest assumption: *price-taking firms*
- this means that each firm **takes the market price as given, and expects to be able to sell as much as it likes at this price**
- this setting is also known as *perfect competition*, and is a good approximation to any situation with:
 - a large number of small firms
 - free entry and exit
 - no transactions costs, prices easily observed

Price-Taking Firms: Example

- suppose one of you decides to start selling pencils
- you are such a small part of global supply, it's safe to expect that you can sell as many as you like, so long as you set the same price as everyone else
- and you are such a small part of global supply, you don't expect to influence the market price

Other Settings We'll Consider Later

- a monopolist looks at the demand curve to figure out the price at which he can sell a particular quantity
- so he does not simply *take* the price as given (i.e. is not a price-taker!), but views it as a function of his own production decisions
- similarly, with a small # firms, the price depends on the firms' collective production decisions, via the demand curve (we'll explore this in game theory)
- but with a large # small firms, each individual firm thinks he's too small to influence the market price, and so takes it as given

Price-Taking Firms: Profits

- suppose a price-taking firm enters when the market price is p
- what he gets to *choose* is his production quantity, q
- let $C(q)$ be the total cost of producing q units
 - we'll derive this from production technology later today, and break it down into different types of costs
- we'll call $MC(q) \equiv C'(q)$ the *marginal cost*
- the firm's goal is to maximize profits, $\Pi(q)$:

$$\Pi(q) = \underbrace{p \cdot q}_{\text{revenue } R(q)} - C(q)$$

Supply with Price-Taking Firms

Step 1: The FOC

- the firm gets to choose q , and wants to maximize

$$\Pi(q) = p \cdot q - C(q)$$

- the optimal (profit-maximizing) q solves the FOC:

$$0 = \Pi'(q) = p - MC$$

$$\Rightarrow \boxed{p=MC}$$

Step 2: The SOC

- we need to make sure that the FOC yields a profit-**maximizing** (vs minimizing) quantity
- this needs:

$$\begin{aligned} 0 &\geq \Pi''(q) \\ &= \frac{d}{dq} \left(\underbrace{p - MC}_{\Pi'(q)} \right) \\ &\iff -\frac{d}{dq} MC \leq 0, \text{ i.e. } \frac{d}{dq} MC \geq 0 \\ &\iff \boxed{MC \uparrow \text{ in } q} \end{aligned}$$

- otherwise, the firm goes for a corner solution

Step 3: Shut-Down Decisions

- finally, let q^* be the quantity that solves $p = MC$ (the FOC), and suppose the SOC holds
- the firm still needs to make sure that operating beats shutting down
- this is the case so long as it is **possible** to choose a quantity where revenue exceeds *avoidable* costs
 - e.g. for a restaurant, employee wages are likely an avoidable cost even in the short run, whereas leasing the restaurant space and manufacturing equipment are not—so view them as “sunk” and factor out of SR decisions
- can boil this down to a very simple condition: operate in SR only if

$p \geq \min AVC$

 - if $p > \min AVC$, go ahead and maximize profits: this means it's possible to produce where each unit brings in more than it costs to make, fixed costs aside
 - if $p < \min AVC$, then, even fixed costs aside, you'll lose money no matter what you produce: shut down!

Sidebar: Types of Costs

- the total cost, $C(q)$, can be broken down into the sum of two costs:
 - fixed costs (F): these do not vary with the production quantity
 - variable costs ($VC(q)$): these depend on production quantity q
- the marginal cost, important from the FOC, is simply $MC \equiv C'(q)$
- *average variable cost [of producing q units]* is $AVC = VC/q$

Sidebar: Calculating Min AVC

- in many cases, it is obviously zero, achieved by $q = 0$; but if not,
- Step (i): set $AVC = MC$, and solve for q
 - this is the quantity, q , that minimizes AVC
- Step (ii) plug the q from Step (i) into AVC (or equivalently, by their equality, into MC)
 - this is the minimum value (across all quantities) of the AVC

Sidebar: Derivation of Min AVC Calculation

- the q that minimizes AVC solves the FOC:

$$\begin{aligned}0 &= AVC'(q) \equiv \frac{d}{dq} \frac{VC(q)}{q} \\&\Rightarrow \frac{qVC'(q) - VC(q)}{q^2} = 0 \\&\Rightarrow qMC - VC = 0 \\&\Rightarrow MC = \frac{VC}{q} \equiv AVC\end{aligned}$$

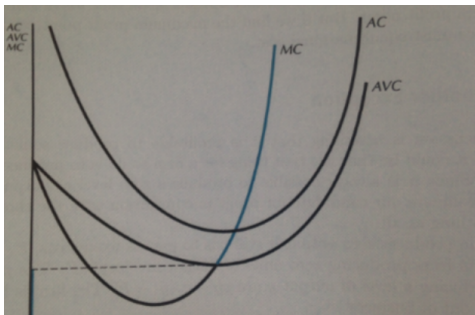
- i.e., it is the quantity where $MC = AVC$
- note for 3rd line: $MC = C'(q) = VC'(q)$, since $C(q) = VC(q) + \text{fixed costs}$

Summary: SR Supply w/ Perfect Competition

So in practice, to find each firm's supply curve in a perfectly competitive setting,

- first, make sure that marginal costs are increasing in quantity (if not, you need to look at corner solutions!)
- if yes, then make sure $p \geq \min_q AVC$; if not, the firm should shut down
 - to get $\min_q AVC$, set $AVC = MC$, solve for q , and evaluate AVC at this q
- if yes, then the firm should produce where $p = MC$
- this is sometimes described as: “the SR supply curve is the part of the MC curve above AVC”

SR Supply in a Picture (Blue)



Example

A firm has fixed costs $F = 1$, and variable costs $VC(q) = q^2$ if it produces q units of output. The firm is a price-taker, and can sell as much as it wants at the market price p .

Find the firm's SR supply curve, and profits.

Solution to Example

- we know that the supply curve is given by $p = MC$ (Step 1 FOC), so long as two conditions are satisfied:
 - (Step 3) operating in the SR beats shutdown, i.e. $p \geq \min_q AVC(q)$
 - (Step 2) and the SOC holds: MC is increasing in q

Step 3: SR Shutdown Decisions

- let's check condition (1) first: we know AVC is minimized where $MC = AVC$; here, recalling $VC(q) = q^2$,

$$\begin{aligned} MC &= 2q, \text{ and } AVC = \frac{q^2}{q} = q \\ \text{so } MC &= AVC \Rightarrow 2q = q \Rightarrow q = 0 \\ &\Rightarrow \min AVC = [AVC]_{q=0} = 0 \end{aligned}$$

- so, this firm will produce as long as the market price is nonnegative (i.e. always)
- you could have also skipped the math in this example: clearly, with $AVC = q$, the lowest possible AVC is zero, achieved by $q = 0$

Step 2: Checking the SOC

- next, let's check condition (2): is $MC \uparrow$ in q ?
- yere, $MC = C'(q) = 2q$
- yes, clearly this is increasing in q (either eyeball it or verify that $MC'(q) = C''(q) > 0$)

Step 1: The FOC

- thus, the SOC and “operating beats shutting down” conditions both hold for any price $p \geq 0$
- so the SR supply curve is simply $p = MC$; here $MC = 2q$, so we get the SR supply curve

$$p = 2q, \text{ or } q = \frac{p}{2}$$

- the first arrangement $p = 2q$ is the *SR inverse supply curve*, and tells us that in order to want to sell q units of output, the firm must get $\$2q$ per unit
- the second arrangement $q = \frac{p}{2}$ is the *SR supply curve*, and tells us that if the market price is p , the firm will produce $\frac{p}{2}$ units of the good

Next: Deriving Cost Functions (?)

- so far, we simply assumed that each firm knew its cost curve $C(q)$, i.e. the cost of producing q units of its product
- but next, we can derive this if we know the firm's *production technology*
- a typical production function describes how inputs are transformed into outputs
- for example, if it takes capital and labor to produce a good, a typical production function is $q = f(K, L)$: the # units you produce (q) is a function (f) of your # units capital (K) and labor (L)

SR Cost Functions

- in the short run, we typically assume capital is fixed ($K = \bar{K}$) but labor can be chosen
- with only one input that the firm may choose, it's simple to get costs in terms of q :
 - use the production function to figure out how many labor units $L(q)$ are needed to produce q output units
 - then calculate the corresponding labor cost!
 - adding fixed capital costs, this gives the cost to produce q units of output
- so, $C^{SR}(q) = r\bar{K} + wL(q)$

LR Cost Functions

- in the long run, both capital and labor (maybe also other inputs) may be chosen
- if the firm gets to choose multiple inputs, then there are typically multiple ways to produce q units
 - profit-maximizing firms want to keep their costs as low as possible
 - they do this by choosing the *cheapest* input combination that produces q units
- formally, $C(q)$ is the cost of the input bundle that most efficiently produces q units
- if the inputs are capital and labor, with respective prices r and w ,

$$C(q) = \min_{\{(K,L):f(K,L)=q\}} (rK + wL)$$

Finding LR Cost Functions

- option #1: use a Lagrangian: minimize $rK + wL$ s.t. $f(K, L) = q$
- option #2 (intuitive method): find optimal input bundle by solving $\frac{MP_K}{MP_L} = \frac{r}{w}$ and $f(K, L) = q$ (for an interior solution)
 - then if you get the bundle $(K^*(q), L^*(q))$, its cost is $C^{LR} = rK^*(q) + wL^*(q)$
 - notation note: $MP_K = \partial f / \partial K$ and $MP_L = \partial f / \partial L$, these are “marginal products”
- this is the production analog to finding the compensated demand bundle and expenditure function
- instead of finding the bundle that most cheaply achieves a utility target, and then its cost, we want the bundle that most cheaply achieves a production target, and then its cost

Example: SR Cost Function

- a firm produces pencils using capital and labor, according to the production function $q = K^{\frac{1}{2}}L^{\frac{1}{2}}$
- however, in the short run, capital is fixed at $K = 100$
- what is the firm's SR cost of producing q units, $C^{\text{SR}}(q)$, if each capital unit costs r and each labor unit costs w ?

Example: SR Cost Function

- answer: here, with K fixed at 100, the firm's production function is $q = 100^{\frac{1}{2}} L^{\frac{1}{2}} = 10\sqrt{L}$
- so the firm only gets to choose labor
- to figure out how many labor units are needed to produce q pencils, simply invert the production function:

$$q = 10\sqrt{L} \Rightarrow L = \frac{q^2}{100}$$

- now, $C^{SR}(q)$ is the cost of this input bundle $(\bar{K}, L(q)) = (100, \frac{q^2}{100})$:

$$C^{SR}(q) = 100r + \frac{q^2}{100}w$$

Long-Run Supply

- one difference between LR supply and SR supply is that LR supply uses the **LR cost function**
- the other difference between LR and SR supply is the shutdown price

LR Shutdown Price

- in the LR, all costs are avoidable
- so price-taking firms will only operate if it's possible to make nonnegative overall profits
- this is possible iff the market price exceeds the minimum average **total** cost, $\min_q AC$
 - to find the q that minimizes AC , set $AC = MC$ and solve for q
 - to find $\min_q AC$, substitute this q back into AC

Market Supply

- to derive market supply, we add up the individual market supply curves across firms
- the market supply curve, $Q^S(p)$, tells you the total quantity (Q) supplied at any price p
- if you “invert” the equation, solving for p in terms of Q , you get the market inverse supply curve $p^S(Q)$, telling you the price at which the market is willing to supply Q units of the good

SR Equilibrium

- in equilibrium, **demand equals supply**
- equilibrium price: the price p^* at which

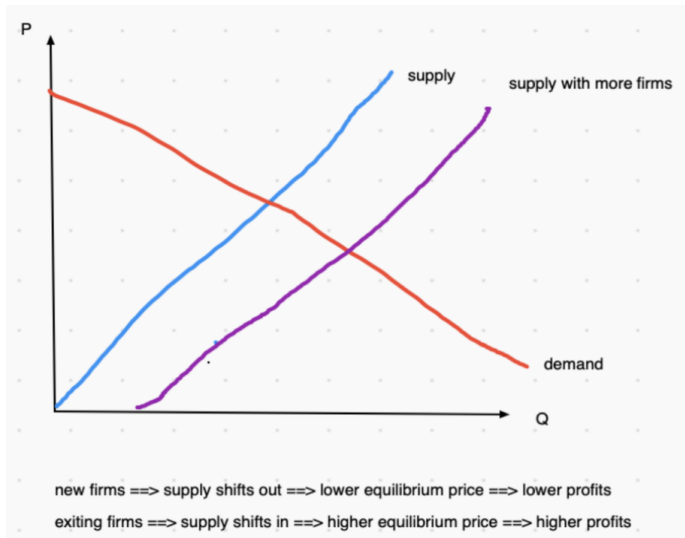
$$Q^S(p^*) = Q^D(p^*)$$

- equilibrium quantity demanded/supplied: $Q^S(p^*) = Q^D(p^*)$
- note: this is **ALWAYS** an equilibrium requirement (whether LR or SR)
- but it is the **ONLY** requirement for a SR equilibrium, while LR equilibrium requires other things too

Long Run Equilibrium: Motivation

- in the long run, we can pin down exactly the # firms and the equilibrium price:
- if firms currently in the market are making money, new firms will come in, shifting supply to the right, thus decreasing the SR equilibrium price
- if firms currently in the market are losing money, some of them will exit, shifting supply to the left, thus increasing the SR equilibrium price

Long Run Equilibrium: Dynamics



Long Run Equilibrium: Dynamics

- but, an increase in the SR equilibrium price increases profits for all firms
- and a decrease in the SR equilibrium price decreases profits for all firms
- so if profits are positive, new firms enter, reducing the equilibrium price, and thus reducing profits
- and if profits are negative, some firms will exit, boosting the equilibrium price, and thus increasing profits
- assuming free entry, these adjustments will keep happening until no firms want to enter or exit
- but when does this happen?

Long Run Equilibrium

- **only when profits are zero for all firms!**
 - caveat: this assumes that all firms have identical cost functions
 - if some firms are more efficient than others, we only need the *least-efficient* surviving firms to break even (example coming up)

Long Run Equilibrium Condition

- thus, in perfect competition with identical firms, **all firms make zero profits**
- this requires that $p = \min AC$:
 - if $p < \min AC$, then each unit sells for a lower price than it can possibly (on average) cost to make, so profits are negative, and this can't be a LR equilibrium (some firms would exit)
 - if $p > \min AC$, then it is *possible* for firms to make positive profits, also not a LR equilibrium (some firms would enter)
- of course, firms are still profit-maximizing, so we also need $p = MC$
- this pins down uniquely the long-run equilibrium price, without even having to look at market demand:

LR equilibrium price p^* : where $p^* = MC = AC$

How To Use This Equation

- set $MC = AC$ and solve for q
- this is each firm's long-run equilibrium supply quantity, call it q^*
- then set $p^* = [AC]_{q=q^*}$ or equivalently $p^* = [MC]_{q=q^*}$
- this is the long-run equilibrium price
- we only need the demand curve to figure out the equilibrium # firms

LR Equilibrium # Firms

- in an equilibrium, we still need demand = supply
- from the demand curve, find the total quantity demanded at equilibrium price p^* , i.e. $Q^D(p^*)$
- if each firm supplies q^* , then N firms supply $N \cdot q^*$ (this is total market supply)
- so equilibrium requires:

$$Nq^* = Q^D(p^*)$$

Example 1

- each firm has cost function $C(q) = q^2 + 1$
- market demand is $Q^D(p) = 100 - 5p$
- find the SR equilibrium price, assuming 10 price-taking firms
- find the LR equilibrium price, quantity, and # firms

Example 1: SR Equilibrium

- to find the SR shutdown price,

$$\begin{aligned} MC &= AVC \Rightarrow 2q = q \Rightarrow q = 0 \\ &\Rightarrow \min AVC = [AVC]_{q=0} = 0 \end{aligned}$$

- so since $MC = 2q$ is increasing in q (thus SOC holds), each firm produces where

$$p = MC \Rightarrow p = 2q \Rightarrow q = \frac{p}{2}$$

- so at price p , 10 price-taking firms produce $10 \cdot \frac{p}{2} = 5p$
- so the equilibrium price sets

$$\text{demand} = \text{supply} \Rightarrow 100 - 5p = 5p \Rightarrow p = 10$$

LR Equilibrium

- to find the LR equilibrium quantity per firm,

$$AC = MC \Rightarrow q + \frac{1}{q} = 2q \Rightarrow q^* = 1$$

- to find the LR equilibrium price,

$$p^* = [AC]_{q=1} = 2, \text{ or } p^* = [MC]_{q=1} = 2$$

- to find the LR equilibrium # firms,

$$Nq^* = Q^D(p^*) \Rightarrow N \cdot 1 = 100 - 5(2) \Rightarrow N = 90$$

Example 2

- consider a market of infinitesimally small firms
- for every $m > 0$, there are m firms with indices in $[0, m]$ (don't worry about the possibility of non-integer quantities)
- the firm with index $a \in [0, m]$ has cost function $C_a(q) = \frac{a^2}{2} + \frac{q^2}{2}$
- firms are price takers and face demand curve $Q^D(p) = 10 - 3p$
- find the LR shutdown price and supply curve for firm a , and the LR equilibrium price
 - assume fixed costs are escapable in the LR (in precept you'll find the SR supply curves)

Solution

- firm a produces where $p = MC = q$, provided that $p \geq \min AC$:
 - $MC = AC \Rightarrow q = \frac{a^2}{2q} + \frac{q}{2} \Rightarrow q = a$
 - at this point, $\min AC = a$ (so firm a produces where $p = q$ provided that $p \geq a$)
- so at price p , each firm $a \leq p$ (there are p of these) produces $q = p$ units
- so market supply at price p is p^2
- equating with demand, we get a LR equilibrium price of 2

Final Notes on Cost Functions

- last class, we learned how to derive $C(Q)$ from the production function
- in the short run, capital is typically fixed, so $C^{SR}(Q)$ is just the fixed capital cost plus the cost of the labor needed to produce Q
- but this short-run (fixed-capital) production technology typically exhibits *eventually diminishing returns*:
 - at least eventually (for high L), the marginal product of labor, $MP_L \equiv \partial f / \partial L$, is decreasing in L
 - this means that eventually, each worker is less and less useful for production
 - “too many cooks in the kitchen”
- this is what leads to an increasing MC , our second-order condition for profit maximization
- if workers are increasingly useless, then it becomes more and more difficult (increasingly costly) to ramp up production

New Topic: Imperfect Competition

- perfect competition means a large # firms, where each thinks he is too small for his quantity chosen to affect the market price
- we ended up with the depressing result that nobody will make any money
- this is *not* the case with a small # firms, known as “imperfect competition”:
 - monopoly: 1 firm
 - duopoly: 2 firms
 - oligopoly: some (small) number n firms

Monopoly

Sometimes, there may be only one supplier

- e.g. electricity, cable (depending on where you live)
- e.g. Apple effectively had a monopoly on tablets when it first introduced the ipad, due to the simple fact that there were no comparable products on the market (and entry was difficult)
- they are surely making money, and it is difficult-to-impossible for other competitors to enter
- a monopoly might arise due to government regulations, or because the monopolist has superior technology or access to essential ingredients that makes his costs much lower than competitors, or because he is able to block other firms from entering, or...

Monopoly

Compared to what we've been doing:

- a monopolist, as with a perfectly competitive firm, wants to maximize profits = revenue - cost
- the *difference*:
 - with perfect competition, each firm believes he is too small to affect market price
 - a monopolist knows that he is the only supplier, and looks at the market demand curve to determine the relationship between the quantity he sells, and the price at which he can sell it

Example

- consider a monopolist, who can produce q units at total cost $C(Q) = Q^2 + 1$, in a market with inverse demand curve $P^D(Q) = 400 - Q$
- our monopolist wants to maximize profits:

$$\begin{aligned}\Pi(q) &= QP^D(Q) - C(Q) \\ &= Q(400 - Q) - (Q^2 + 1)\end{aligned}$$

- first order condition for profit maximization:

$$\begin{aligned}0 &= \Pi'(Q) = Q(-1) + (400 - Q) - 2Q \\ &= 400 - 4Q \\ \Rightarrow Q &= 100\end{aligned}$$

Example Cont'd

- second order condition for profit maximization is $\Pi''(Q) < 0$
- here, $\Pi'(Q) = 100 - 4Q$, so $\Pi''(Q) = -4$
- this is indeed negative, so the first-order condition is indeed finding a max, not a min

Difference from Perfect Competition

- both want to choose their quantity to maximize profits = revenue - cost
- both do this via a FOC:

$$0 = \Pi'(Q) = MR - MC$$

where $MR \equiv R'(Q)$ (“marginal revenue”)

- but a perfectly competitive firm takes p as given and constant: so

$$R(Q) = pQ \Rightarrow MR = p$$

- a monopolist is the entire market, so price = $P^D(Q)$ is the demand price (depends on Q):

$$R(Q) = P^D(Q) \cdot Q \Rightarrow MR = P^D(Q) + Q \frac{d}{dQ} P^D(Q)$$

Difference from Perfect Competition

- assuming downward-sloping demand, $\frac{d}{dQ} P^D(Q) < 0$
- so a monopolist's marginal revenue is smaller than the price
- intuition: marginal revenue means the boost in revenue from selling the next unit
- for a perfectly competitive firm, this is simply the price that it brings in
- for the monopolist, it is less, since selling more also reduces the price at which all units can be sold!
- monopolists typically produce less than perfectly competitive firms, due to the incentive to cut back on quantity so as to boost the market price